A 1.1.7 Secondary Memory

A 1.1.7 Describe internal and external types of secondary memory storage

Primary memory loses its contents the moment the power goes off, so every computer needs somewhere permanent to keep its operating system, applications and files. That is the job of secondary memory: non-volatile storage that holds data whether the machine is running or not. The syllabus divides secondary storage into two types, internal storage fitted inside the computer case and external storage connected from outside it. This page describes the internal types (HDD, SSD and eMMC), the external types (external drives, optical drives, flash drives, memory cards and NAS), and the differences between the two groups.

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Key Questions

- What distinguishes internal from external secondary memory storage?
- What are the characteristics and typical uses of HDD, SSD and eMMC as internal storage?
- How do external HDDs and SSDs, optical drives, flash drives, memory cards and NAS differ in their applications?

IB Command Term note

The command term here is ***describe***: give a detailed account.

Naming a device is not describing it. For each storage type you should be able to state the technology it uses, one or two defining characteristics, and a typical use. “An SSD stores data in flash memory, has no moving parts, and is used to hold the operating system because it gives fast boot times” is a description; “an SSD is a fast drive” is not. You are not asked to evaluate or compare beyond the internal and external distinction the statement names.

What separates internal from external storage?

The distinction is about location and connection, not about what the device stores. Internal storage sits inside the computer case and connects directly to the motherboard, which is why it is usually faster: data does not have to pass through an intermediate interface. External storage sits outside the case and connects through an interface such as USB, or over a network. That interface is the bottleneck, so access is generally slower, but in exchange the device becomes portable. You can unplug an external drive and carry your files to another machine; you cannot do that with the SSD soldered inside a laptop.

	Internal storage	External storage
Location	Inside the computer case	Outside the computer case
Connection	Direct to the motherboard	Via an interface (USB, network)
Typical speed	Faster	Slower, limited by the interface
Portability	Fixed in place	Portable between machines
Examples	HDD, SSD, eMMC	External HDD/SSD, optical drive, flash drive, memory card, NAS

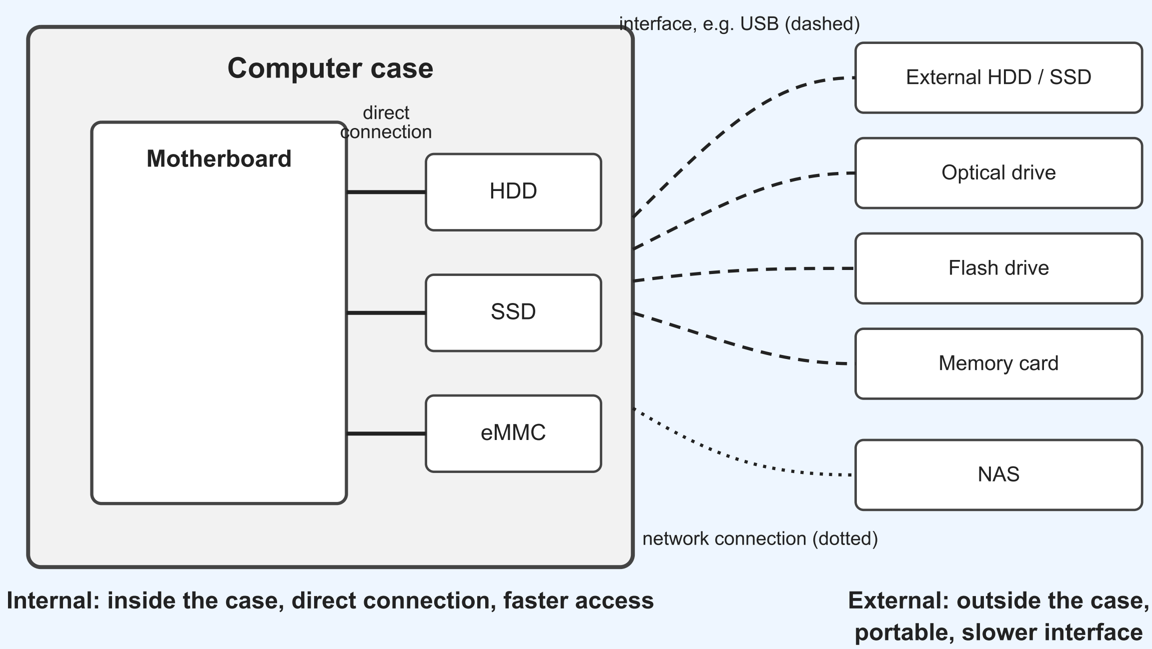

Figure 1. Internal storage connects directly to the motherboard; external storage connects through an interface such as USB or the network.

Internal secondary storage

THINK FIRST

Your phone has no drive bay and nothing plugged into it, yet it keeps thousands of photos safe through every restart. What kind of storage must it contain, and is that storage internal or external?

HDD (Hard Disk Drive)

An HDD stores data magnetically on spinning platters, read and written by a moving head. Because it has mechanical parts, access is slower than flash-based storage and the drive is more vulnerable to damage if dropped, but it offers the lowest cost per gigabyte of any internal option. The consequence is that HDDs suit situations where capacity matters more than speed: operating systems on budget machines, and large collections of files such as video and music.

A school server holding years of student coursework and CCTV footage uses HDDs because terabytes of capacity are needed and millisecond access times are not.

SSD (Solid State Drive)

An SSD stores data in flash memory with no moving parts at all. That single design difference explains everything else about it: access is significantly faster, the drive survives knocks that would destroy an HDD, and it costs more per gigabyte. SSDs are the standard choice for holding the operating system and applications, because the speed of the drive is what the user feels every time the machine boots or a program loads.

A laptop with its operating system on an SSD boots in seconds; the same machine with an HDD takes noticeably longer to reach the desktop.

eMMC (Embedded MultiMediaCard)

eMMC is flash memory soldered directly onto a device's circuit board rather than fitted as a separate drive. It is slower and less durable than a full SSD, but it is compact and cheap, which is exactly what a small device needs. You will find it as the built-in storage in phones, tablets and budget laptops, holding the operating system and the user's apps and files.

A £150 tablet stores its operating system and apps on 64 GB of eMMC because there is no physical room for a separate drive and the price would not allow one.

External secondary storage

External devices use much the same underlying technologies, magnetic and flash, but packaged for connection from outside the case. What differs between them is the application each is designed for.

External HDD and SSD

These are the same drive technologies described above, enclosed in a casing and connected through a port such as USB. The drive itself may be quick, but the interface limits the speed, so they are chosen for portability rather than performance. Typical uses are backup, moving large amounts of data between machines, and extending the storage of a computer whose internal drive is full.

A student backs up two years of Internal Assessment work to an external SSD at the end of every week, so a stolen laptop does not mean a lost project.

Other Storage

Optical drives (CD, DVD, Blu-ray)

Optical drives use a laser to read and write data as microscopic marks on a disc. Access is slow by modern standards and capacities are modest, but the discs themselves are cheap to produce in quantity, which kept optical media in use for software distribution, film and music playback, and long-term archiving.

A games console reads a Blu-ray disc to install a game; the disc also acts as proof of purchase and a permanent archive copy.

Flash drives (USB drives)

A flash drive is a small amount of flash memory behind a USB connector. Its value is convenience: it fits on a keyring, needs no cable and works in almost any computer. It is used for portable storage, transferring files between machines, and booting an operating system for installation or repair.

A teacher carries the day's presentations on a flash drive and plugs it into whichever classroom machine is available.

Memory cards (SD, microSD)

Memory cards are flash storage in an even smaller format, designed to slot inside portable devices rather than hang off a port. Their application is expanding the storage of cameras, phones, drones and handheld consoles, typically to hold photos and video that would quickly fill the device's built-in storage.

A photographer fits a 128 GB microSD card into a camera to hold a full day of high-resolution images.

NAS (Network Attached Storage)

A NAS is a dedicated storage device connected to a network rather than to one computer. It provides centralised storage that every device on the network can reach, which makes it the odd one out in this list: it is external storage shared by many machines at once. Typical uses are file sharing, automated backups and media streaming around a home or organisation.

A Computer Science department stores all student project folders on a NAS, so the same files are reachable from any machine in either lab.

COMMON EXAM MISTAKE

Writing that external storage is volatile, or treating "external" as meaning "less permanent". All secondary storage, internal and external, is non-volatile: the data survives power-off.

The internal and external distinction is about location and connection, and a common giveaway of confusion when first meeting this topic is an answer that mixes it up with the primary versus secondary distinction.

Primary versus secondary is about volatility and purpose. Internal versus external is about where the secondary device sits and how it connects.

Summary Reference

Storage type	Internal or external	Technology	Typical use
HDD	Internal	Magnetic platters, moving parts	OS and large files where capacity beats speed
SSD	Internal	Flash memory, no moving parts	OS and applications where speed matters
eMMC	Internal	Flash memory soldered to the board	Built-in storage in phones, tablets, budget laptops
External HDD/SSD	External	Magnetic or flash, in an enclosure	Backup, portable storage, extra capacity
Optical drive	External	Laser reads marks on a disc	Software distribution, media playback, archiving
Flash drive	External	Flash memory with USB connector	Transferring files, portable storage, booting an OS
Memory card	External	Flash memory, card format	Expanding storage in cameras and portable devices
NAS	External	Networked enclosure of drives	Centralised storage, file sharing, backups, streaming

Try it on paper

The quiz below checks recognition. These two questions practise the written form the command term demands. Answer on paper before checking the model answers.

1. Describe two differences between internal and external secondary memory storage. [2 marks]

Model answer: Internal storage is located inside the computer case and connects directly to the motherboard, giving faster access [1]. External storage connects from outside through an interface such as USB or a network, so it is portable and can be moved between computers but access is generally slower [1].

2. Describe the characteristics of an SSD and state one reason it is preferred to an HDD for storing an operating system. [3 marks]

Model answer: An SSD stores data in flash memory [1]. It has no moving parts, which makes access significantly faster and the drive more durable than an HDD [1]. It is preferred for the operating system because its faster access gives quicker boot and program loading times [1].

A 1.1.7 Quiz

1

Which of the following is a key difference between internal and external secondary memory?

A. Internal memory has a larger storage capacity than external memory.

B. Internal memory is volatile, while external memory is non-volatile.

C. Internal memory is more portable and easily swapped between devices.

D. Internal memory is typically faster due to a direct connection to the motherboard.

2

Which type of internal storage uses magnetic platters to store data?

A. SSD

B. Flash drive

C. HDD

D. eMMC

3

What is a key advantage of SSDs over HDDs?

A. SSDs have faster access speeds due to no moving parts.

B. SSDs have a larger storage capacity.

C. SSDs consume more power.

D. SSDs are less expensive.

4

Which type of storage is often found in small devices like tablets and smartphones due to its compact size?

A. HDD

B. Optical drive

C. eMMC

D. NAS

5

Which of the following is NOT an example of an optical drive?

A. Blu-ray Disc

B. DVD-RW

C. Flash drive

D. CD-ROM

6

What is the primary use of a flash drive?

A. To store the operating system.

B. To provide portable storage and transfer files between computers.

C. To permanently store large media files.

D. To back up an entire computer system.

7

What is a key advantage of using a NAS (Network Attached Storage) device?

A. It is the most portable type of storage.

B. It provides centralized storage accessible by multiple devices over a network.

C. It is faster than an internal SSD.

D. It uses optical discs for long-term archiving.

8

Which type of storage is MOST suitable for expanding the storage capacity of a smartphone?

A. Optical drive

B. NAS

C. External HDD

D. Memory card

9

What is a common misconception about secondary storage?

A. Secondary storage is volatile, meaning it loses data when the power is turned off.

B. All types of secondary storage have the same access speeds.

C. Cloud storage is a type of internal secondary storage.

D. Secondary storage is only used for storing personal files.

10

Why is it important to understand the differences between various types of secondary storage?

A. To make informed decisions about choosing the right storage for different needs and app

B. To be able to repair and replace storage devices.

C. It is not important for most computer users.

D. To impress friends with technical knowledge.